

MULLINGAR, Ireland

WMO: 039710

Lat: 53.533N

Lon: 7.367W

Elev: 328

StdP: 14.52

Time Zone: 1.00 (IRL)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 25.0	(c) 28.0	(d) 22.4	(e) 17.1	(f) 28.4	(g) 25.2	(h) 19.5	(i) 31.2	(j) 24.4	(k) 48.3	(l) 22.1	(m) 48.1	(n) 2.6	(o) 350	(p) 0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	72.6	62.9	69.7	61.5	67.2	60.2	64.6	69.7	62.9	67.4	61.5	65.7	7.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	85.8	66.8	61.0	81.2	65.2	59.6	77.0	63.6	29.8	70.0	28.6	67.5	27.6	65.8	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.0	(b) 17.6	(c) 15.6	(e) 20.5	(f) 78.2	(g) 3.7	(h) 3.2	(i) 17.8	(j) 80.5	(k) 15.6	(l) 82.4	(m) 13.5	(n) 84.2	(o) 10.8	(p) 86.5		
(a) 20.0	(b) 17.6	(c) 15.6	(e) 20.0	(f) 67.9	(g) 3.8	(h) 2.1	(i) 17.3	(j) 69.4	(k) 15.2	(l) 70.7	(m) 13.0	(n) 71.9	(o) 10.3	(p) 73.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	40.7	41.2	43.2	46.7	51.7	56.2	59.1	58.3	55.2	49.8	44.2	41.1
	DBStd	8.20	5.22	5.03	4.91	4.57	4.39	4.29	3.83	3.44	4.09	4.83	5.03	6.15
	HDD50	1432	289	249	215	116	31	3	0	0	5	63	182	279
	HDD65	5846	753	667	675	548	412	264	189	209	294	470	623	742
	CDD50	1070	1	2	5	19	84	190	281	258	162	58	8	2
	CDD65	8	0	0	0	0	0	1	5	2	0	0	0	0
	CDH74	65	0	0	0	0	1	17	39	8	0	0	0	0
	CDH80	4	0	0	0	0	0	1	2	1	0	0	0	0
Wind	WSAvg	7.6	8.6	8.4	8.1	7.8	7.5	6.7	6.4	6.6	7.0	7.6	7.9	8.4
Precipitation	PrecAvg	37.2	3.7	2.8	2.8	2.5	2.7	2.6	2.6	3.2	3.3	3.7	3.6	3.7
	PrecMax	46.0	6.1	7.6	5.0	4.4	5.8	5.5	6.2	7.1	7.2	6.8	8.0	6.4
	PrecMin	25.5	1.1	0.4	0.6	0.8	0.5	0.7	0.6	0.3	0.1	1.4	1.4	0.8
	PrecStd	4.3	1.4	1.5	1.1	1.0	1.4	1.3	1.2	1.4	1.6	1.5	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	55.5	60.1	65.2	71.8	76.2	78.8	73.6	70.7	63.3	58.1	55.7
		MCWB	52.1	52.0	51.4	55.4	61.2	62.8	66.6	64.3	63.0	58.7	55.8	53.4
	2%	DB	52.2	52.8	55.6	61.0	67.4	71.0	73.7	69.7	67.2	60.6	55.5	53.5
		MCWB	50.4	50.2	50.0	52.7	58.4	61.1	63.8	62.5	61.1	57.1	53.6	51.7
	5%	DB	50.7	51.1	53.2	57.6	63.4	68.0	69.8	67.1	64.7	59.0	53.9	52.0
		MCWB	49.0	48.8	49.1	50.9	55.8	59.7	62.0	60.8	59.4	56.0	52.2	50.4
	10%	DB	49.2	49.5	51.3	54.9	60.3	64.9	66.7	65.0	62.5	57.5	52.3	50.2
		MCWB	47.5	47.3	47.9	49.6	54.0	58.0	60.1	59.7	58.0	54.7	50.5	48.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	52.6	53.2	56.7	62.6	64.7	67.7	66.4	64.8	60.2	56.4	53.9
		MCDB	53.6	54.3	57.4	63.7	70.5	71.8	76.8	70.7	68.4	62.0	57.4	55.0
	2%	WB	50.9	50.9	51.5	54.0	59.3	62.4	65.1	63.8	62.4	58.2	54.1	52.0
		MCDB	52.0	52.5	54.2	59.0	66.2	69.1	71.3	67.4	65.4	59.8	55.1	53.1
	5%	WB	49.4	49.3	50.1	52.1	56.9	60.5	62.9	62.4	60.5	56.7	52.5	50.6
		MCDB	50.6	50.9	52.7	56.3	61.7	66.4	67.9	65.6	63.5	58.5	53.7	51.8
	10%	WB	47.7	47.6	48.3	50.2	54.9	58.8	61.2	60.8	58.9	55.1	50.8	48.7
		MCDB	49.0	49.3	50.9	54.1	59.1	63.9	65.5	63.9	61.5	57.0	52.2	50.0
Mean Daily Temperature Range	5% DB	MDBR	9.7	11.0	13.0	14.2	14.6	14.5	13.8	13.5	13.3	12.2	10.6	9.8
		MCDBR	10.7	10.7	14.6	18.5	19.7	19.4	18.7	15.8	15.4	11.9	10.6	10.7
		MCWBR	10.2	9.2	10.1	11.5	11.9	11.1	10.5	9.5	10.0	9.2	9.6	10.3
	5% WB	MCDBR	10.7	10.7	13.0	16.3	17.2	17.4	16.3	13.7	13.4	10.7	10.2	10.6
		MCWBR	10.4	9.8	10.1	10.8	11.1	10.7	10.3	9.5	9.9	9.2	9.8	10.4
Clear-Sky Solar Irradiance	taub	0.332	0.342	0.365	0.375	0.384	0.373	0.376	0.358	0.352	0.340	0.326	0.322	
	taud	2.247	2.316	2.280	2.281	2.292	2.351	2.350	2.411	2.427	2.415	2.306	2.211	
	Ebn at Noon	194	237	257	271	274	278	275	274	260	236	197	170	
	Edh at Noon	20	26	33	37	39	37	36	32	28	23	19	17	
All-Sky Solar Radiation	RadAvg	220	441	783	1198	1521	1574	1425	1197	898	532	278	169	
	RadStd	20	38	83	126	116	167	130	113	46	43	30	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page